

Vishal Bakshi

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Professional Summary

Machine Learning Engineer with 3+ years of experience designing and developing scalable backend services and data pipelines for large-scale AI and LLM applications. Proven expertise in the end-to-end ML lifecycle, from building ETL pipelines for complex datasets to deploying and optimizing models via robust APIs. Proficient in Python and Go, with hands-on experience in MLOps (Docker, Kubernetes, CI/CD) and cloud platforms (AWS).

Work Experience

Data Engineer Intern

Jun 2025 – Present

FCCI Insurance Group, Sarasota, FL

- Maintained core ETL pipelines for critical insurance data, resolving 10+ production bugs to reduce pipeline failures by 95% and ensure on-time delivery of daily financial reports.
- Developed a data validation report using complex SQL joins to proactively identify and flag duplicate claims, preventing an estimated \$20,000 in erroneous payouts per quarter.
- Delivered \$36,000 in business savings by building and deploying a production-grade, multi-threaded Python pipeline to automate legacy code audits using the OpenAI GPT API, reducing manual review effort by 80%.

Machine Learning Engineer

May 2024 – May 2025

HIVE Lab, Washington, DC

- Uncovered that 62.9% of healthy patients exhibited early pre-diabetic patterns by developing an ETL and data workflow for 60 GB of time-series patient data; a critical finding validated with clinical SMEs for early intervention.
- Designed and implemented a model evaluation framework focused on recall for at-risk patient identification; improved performance by 20% through engineering 5+ novel temporal features from complex CGM data.
- Proposed and designed the architecture for a real-time patient feedback system, specifying the RESTful API endpoints and data schema required to reliably translate analytical outputs into actionable health alerts via FastAPI.

Machine Learning Engineer

Jun 2022 – Aug 2023

HBots Robotics Pvt Ltd, Hyderabad, India

- Owned the end-to-end development of a novel computer vision system, from dataset acquisition and model prototyping in PyTorch to production deployment, directly addressing a core business need for automation.
- Architected and built a custom image dataset from the ground up, implementing targeted data augmentation strategies that increased object detection model mAP by 35% and met a critical 3-month product deadline.
- Engineered a sophisticated data fusion pipeline integrating stereo depth cameras with the 2D detection model, enabling precise 3D geometric volume estimation—a key feature for the product's value proposition.

Projects

Intelliwiz - Full-Stack AI Application for Automated Data Intelligence

Jan 2025 – May 2025

Reducing Time Taken to Analyze a Dataset and Come Up with Meaningful Visualization to Understand the Insights

- Developed a production-ready agentic system using LangGraph to automate complex data intelligence workflows, proving the viability of autonomous agents for data processing tasks and reducing manual workload by over 70%.
- Architected an LLM-powered engine to auto-generate 12+ chart types (Plotly-based) from schema analysis, providing immediate visual insights without requiring any user scripting.
- Engineered and deployed the solution as an intuitive Streamlit application, enabling non-technical business stakeholders to generate executive-ready reports in minutes, bridging the gap between raw data and insights.

Transformative NLP Framework for New York Airbnb Review Analysis

Aug 2024 – Dec 2024

Driving Data-Informed Decisions for Enhanced Guest Experiences and Host Performance

- Engineered an advanced NLP pipeline to analyze 17,000+ Airbnb guest reviews, fine-tuning BERT models for sentiment classification, achieving high contextual accuracy to enhance guest experience insights.
- Developed an Aspect-Based Sentiment Analysis system, leveraging KMeans clustering, and aspect tagging to extract granular insights on cleanliness, location, and host interactions, driving data-informed decisions for hosts.
- Designed a transformer-based T5 review summarization model, automating concise, high-quality summaries of guest feedback, reducing manual review time by 60% while improving host responsiveness and service optimization.

Education

The George Washington University, Washington, DC

May 2025

Master of Science in Data Science

GPA: 3.85/4.0

- Awarded the CCAS Dean's Award, an exceptional merit-based scholarship of \$17,000 USD from GWU.

Skills Summary

Programming Languages: Python, Go, Java (Foundational), SQL, R

Backend & MLOps: API Design (FastAPI), Docker, Kubernetes, CI/CD (GitHub Actions), MLFlow, Kafka

AI/ML Frameworks: PyTorch, LangChain, Scikit-learn, Hugging Face, Transformers

Databases & Cloud: PostgreSQL, MySQL, AWS (EC2, SageMaker), GCP, Apache Spark